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Chapter 8

Tips And Techniques



Housekeeping Commands

This section deals with routine and useful commands. The commands are split alphabetically. (C-D-F, M, R-S)

Housekeeping Commands C-D-F

COMPALLPGM - Compress All Programs

Runs the Compress All Programs Routine. This is run by the Batch Job submitted within Execute Day-End Functions, option 7.

DLTUNKLIBS - Delete Unknown Libraries

Runs the Delete Unknown Libraries Routine. This is run by the Batch Job submitted within Execute Day-End Functions, option 5.

DSPDLTAUD - Display Library Deletion Audit

Takes you into the Display the Library Deletions Audit Trail function: select Operational Enquiries, option 5.

DSPDLTRQS - Display Library Delete Request

Takes you into the Display Library Deletion Requests function: select Operational Enquiries, option 4.

DSPMED - Display Media

Takes you into the Display Media function: select Operational Enquiries, option 3.

DSPMYFLR USER(*USER, User ID) - Display User Folders

Takes you to the Display User Folders function. Operational Enquiries, option 2. Parameter USER default value (*USER) shows the Folders Owned under Housekeeping by the Current User, or enter a (User ID) to show the Folders Owned under Housekeeping by that User.

DSPMYLIBS USER(*USER, *SELECT, User ID) CURLIB(0, 1) SLTUSR(0, 1) - Display User Libraries

Takes you to the Display User Libraries function. Operational Enquiries, option 1. Parameter USER default value (*USER) shows the Libraries Owned by the Current User. Value (*SELECT) displays a list of all of the Users registered with Administration Functions for you to select from or enter a (User ID) directly to show the Libraries Owned by that User. Parameter CURLIB default value (0) shows details of ALL of the Libraries Owned by the User, value (1) shows details only of those Libraries Owned by the User that are ON THE SYSTEM this can take A LOT LONGER to respond. Parameter SLTUSR default value (0) takes you in normally, whereas value (1) has the same effect as parameter USER value (*SELECT).

FREFLRMED - Free Folder Save Media

Takes you to the Free Folder Save Media function: select Save to Media, option 7.

FREMED - Free Library Save Media

Takes you to the Free Library Save Media function. Select Save to Media, option 6.

Housekeeping Commands M

MNGARC - Manage Library Archiving Saves

Takes you to the Manage Libraries Archiving function. Select Save to Media, option 2. The Archiving function entered (Select/Mass) is set in the Manage Control Information function.

MNGCON - Manage Control Information

Takes you to the Manage Control Information function. Select Control Details, option 1.

MNGDET TYPE(T, C) - Manage Housekeeping Details

Parameter TYPE default value (T) takes you to the Manage Library Type Codes function. Select Control Details, option 2, whereas value (C) takes you to the Manage Library Contents Codes function. Select Control Details, option 3.

MNGFLR FLR(*SELECT, Folder Path/Name) - Manage Folders

Takes you to the Manage Folders function. Definitions, option 2. Parameter FLR default value (*SELECT) takes you to the Enter Folder Path/Name screen, but if you enter a Folder Path/Name, for example:

(MNGFLR FLR (GENERAL/GENERAL.HK/MANUALS))

you go straight to the Folder Details screen.

MNGLIB LIB(*SELECT, Library Name) - Manage Libraries

Takes you to the Manage Libraries function. Definitions, option 1. Parameter LIB default value (*SELECT) takes you to the Enter Library Name screen, but if you enter a Library Name (for example, MNGLIB LIB(IPGHKP3)) you bypass this and go straight to the Library Details screen.

MNGMED - Manage Library Save Media

Takes you to the Manage Media function. Select Definitions, option 3.

MNGOFF - Manage Library Save Groups

Takes you to the Manage Library Save Groups function. Select Control Details, option 4.

MNGOFFGRP - Manage Save Group To Save

Takes you to the Select Library Save Group to be Saved function. Select Execute Day-End Functions, option 4.

MNGPROLIBS - Manage Protected Libraries

Takes you to the Manage Protected Libraries function. Select Control Details, option 5.

MNGRST - Manage Library Restores

Takes you to the Manage Library Restores function. Select Restore from Media, option 1.

MNGSAV - Manage Library Saves

Takes you to the Manage Library Saves/Backups function. Select Save to Media, option 1. The Save/Backup function entered is set in the Manage Control Information function.

MNGSAVFLR - Manage Folder Saves

Takes you to the Manage Folders Backups function. Select Save to Media, option 3.

MNGSAVOFF - Manage Library Group Saves

Takes you to the Manage Library Save Groups Backups function. Select Save to Media, option 4.

R-S

RGZALLPFMS - Reorganise All Physical Files

Runs the Reorganise All Physical Files Routine. This is run by the Batch Job submitted from Execute Day-End Functions, option 6. This executes the Command RGZLIBPFM for all of the Libraries with the Reorganise Files Flag set in their Housekeeping Details.

RGZLIBPFM LIB(Library Name) - Reorganise Library Files

This executes the IBM AS/400 Command RGZPFM for all of the Physical Files in the Library specified.

RSTHKF - Restore Housekeeping Files

Takes you to the Restore Housekeeping Files function. Select Restore from Media, option 2.

SAVHKF - Save Housekeeping Files

Takes you to the Save Housekeeping Files function. Select Save to Media, option 5.

Housekeeping Data Areas

This area identifies which tasks are assigned to specific data areas.

HKDISKLMT - System Auxiliary Storage Usage Limit For Backups

This holds the Auxiliary Storage Usage level, which you may set, beyond which Auto-backup Saves to Save Files will not take place. For example, if you have set the value to 95% and the Save to Save File of a Library would take the System Auxiliary Storage Usage beyond this level, then the Save will not take place and an appropriate message will be sent to the Message Queue HKSAVERR.

HKFSAVDTA - Last Run Details

This Data Area holds details about the last execution of the Save Housekeeping Files function. That is, of the Off-line Storage Media Volume used.

HKINTFLR - Last Used Internal Folder Name

This Data Area holds the latest Internal Folder name generated by Housekeeping, it is incremented each time that there is a new Folder Registration.

HKOFFGRP - Library Save Group Designated To Be Saved

This Data Area holds the Code of the Library Save Group to be Saved by the Library Save Groups Auto-backups Routine as selected via the Select Library Save Group to be Saved function.

HKSIZDATIM - Size All Libraries Last Run Datetime (Cymmddhhmmss)

This Data Area holds the Date and Time that the last run of the Re-size All Libraries Routine completed.

Housekeeping Message Queues

This section identifies the types of message that appear in specific message queues.

Deletions Message Queue

Messages relating to Housekeeping Library Deletion operations appear on this Message Queue.

HKSAVERR

Housekeeping Save Error Message Queue

If a Library is not Saved to Save File by the Libraries Auto-backups Routine because the System Auxiliary Storage Usage limit would be exceeded, then a message will be sent to this Message Queue.

Housekeeping Auto Day End Routines

This section deals with the routine day end activities and routines that form part of the Housekeeping functionality.

The Libraries Auto Backups Routine

This Routine is run by the Program HK900CLP, which is run by the Batch Job submitted from Execute Day-End Functions, option 1. This Routine clears Work Libraries (Those with Library Type Code W in their Housekeeping details). It attempts to delete libraries flagged for deletion which do not have an outstanding request for a Security Save, and those which have been Archived to Off-line Storage Media.

It also attempts to save the following:

- Save File Libraries which require a Security Save before Deletion.
- Libraries with their Force Single Save Flag set.
- Libraries which are due for Backup according to their Save Frequency and their Force Save flag being set, or changes being made since their Last Save.
- Libraries which have been Flagged for Archiving.

The Folders Auto Backups Routine

This Routine is run by the Program HK970CLP, which is run by the Batch Job submitted from Execute Day-End Functions, option 2. This Routine will attempt to Save:

- Save File Folders with their Force Single Save Flag set.
- Folders which are due for Backup according to their Save Frequency and their Force Save flag being set or changes being made since their Last Save.

The Libraries Save Groups Auto Backups Routine

This Routine is run by the Program HK975CLP, which is run by the Batch Job submitted from Execute Day-End Functions, option 3. This Routine attempts to save libraries assigned to the Save Group selected to be Saved via the Select Library Save Group to be Saved function.

The Size All Libraries Routine

This Routine is run by the Program HK903CLP, which is run by the Batch Job submitted from Execute Day-End Functions, option 8. This Routine will measure the Size of all of the Libraries on the System, this will entail measuring the Size of all of the Objects in all of the Libraries on the System, this may take a very long time to run.

Housekeeping Hints

This section provides some helpful tips to maximise the efficiency of your Housekeeping routines.

Auto Day End Routines

It is recommended that you set up the Housekeeping Auto-Day-End Routines to run under the control of Machine Manager. Since the Size All Libraries Routine can take a long time, run it when the Machine is not busy, for example, at weekends.

Create/Delete Libraries

Since Libraries may be both Created and Deleted via Housekeeping functions, if you wish to use Housekeeping most effectively for Object/Machine (Storage) management, it is recommended that authority to the Commands CRTLIB and DLTLIB be revoked.

Cyclic Normal Backup Method

If you set up Housekeeping to use the Cyclic Normal Backup Method you should set the Housekeeping Details for Libraries to have a Force Save Flag of 1 and a Save Frequency equal to the number of days in the Cycle. This is because this Backup Method will automatically Free the Save Media for the current day in the Cycle regardless of whether or not there are Active Saves on the Media Volumes.

Mass Normal Backup Method

If you set up Housekeeping to use the Mass Normal Backup Method and set Library Force Save Flags to 0, so that Libraries are Backed up according to their Save Frequencies and whether or not they have changed since their last Save, it may be that Libraries that do not change often are therefore not Saved often and Housekeeping will not allow Save Media to be Freed for the sake of one Library's Save. However, you may wish to re-use the Media. You may force a Media Volume to be Freed by setting the Force Single Save Flags of all of the Libraries on that Volume whose Saves are still Active, these Libraries will then be included in the next Backup run and may then be Saved once more, thus the Active Saves will be superseded and become Inactive. When there are no Active Saves on a Media Volume, that Volume may be Freed. This procedure completes over a period of two Backup runs, typically two days.

SAVLIB Failure

The failure of the Libraries Auto-backups Routine to Save a Library to Save File may occur for any of the reasons that may cause the Command SAVLIB to fail. Most commonly this will be due to the User of the Routine not having sufficient authority to the Library, or due to there being Locks held on the Library. When Saves are not made because the given System Auxiliary Storage Usage limit would be exceeded, a message is sent to the Message Queue HKSAVERR.

Save Housekeeping Files

The Save Housekeeping Files operation should be routinely carried out after all of the other Save operations have been completed so that the Data Saved fully records the current state of all of the Saves known to Housekeeping. If the

execution of a Save Housekeeping Files operation is interrupted before it completes normally, there may be copies of the Housekeeping Files, which would normally be Deleted by the operation, left in the Library QTEMP. These should be Deleted or the Job (session) ended before any further Housekeeping operations are carried out, to prevent the copies rather than the real files from being updated.